

Deep Learning

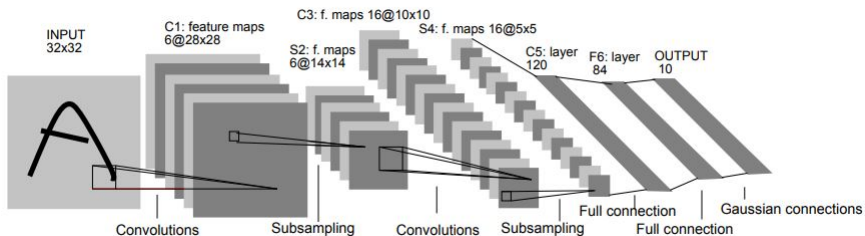
Vazgen Mikayelyan

June 1, 2021

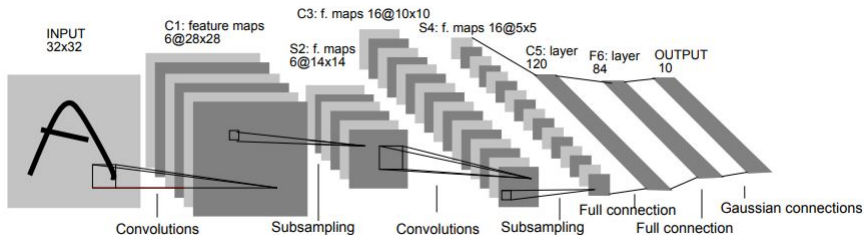


1 Famous CNNs

LeNet-5 (1998)

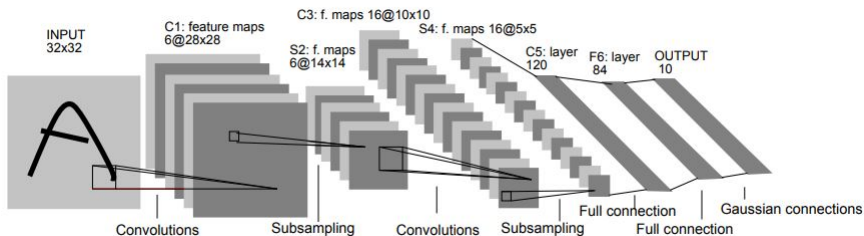


LeNet-5 (1998)



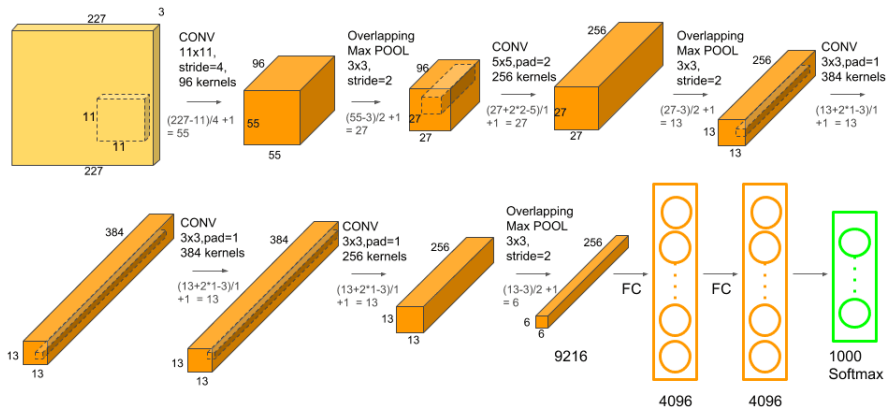
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LeNet-5 (1998)



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- LeNet-5 has approximately 60k parameters.

AlexNet (2012)



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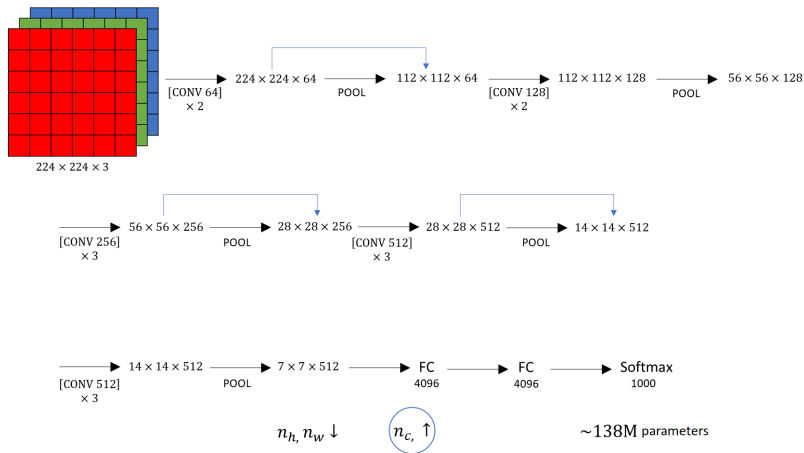
AlexNet (2012)

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- Local response normalization?
- AlexNet has approximately 60M parameters.
- Accuracies on ImageNet: Top1=63.3%, Top5=84.6%.

VGG-16 (2014)

CONV = 3×3 filter, $s=1$, same

MAX-POOL = 2×2 , $s=2$



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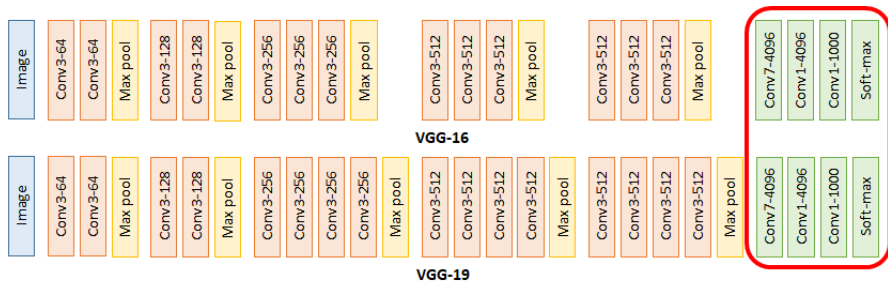
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- VGG-16 has approximately 138M parameters.

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- VGG-16 has approximately 138M parameters.
- Accuracies on ImageNet: Top1=74.4%, Top5=91.9%.

VGG-19 (2014)



- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.

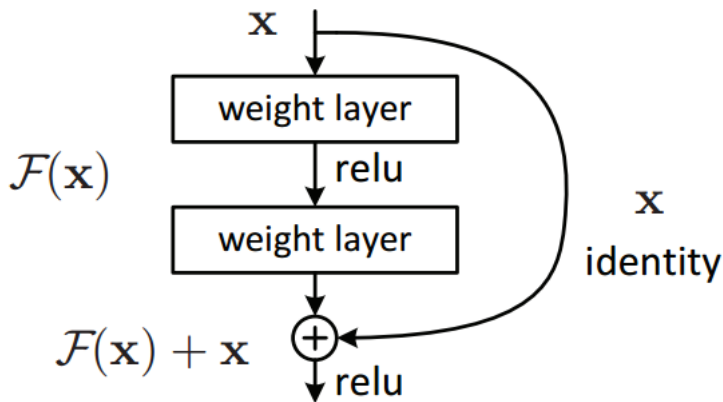
VGG-19 (2014)

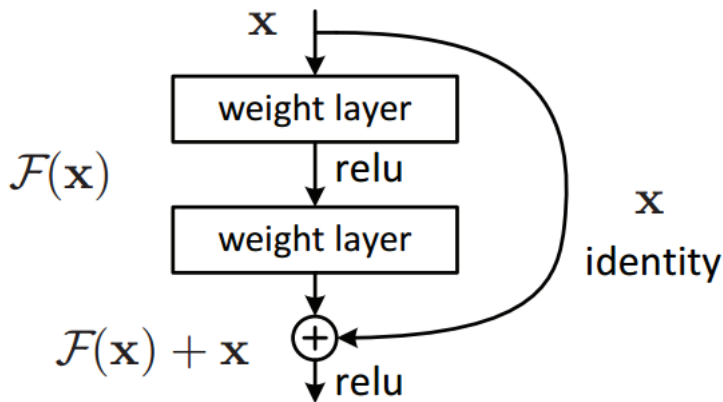
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VGG-19 (2014)

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- VGG-19 has approximately 144M parameters.
- Accuracies on ImageNet: Top1=74.5%, Top5=92%.

Resnet-50 (2015)

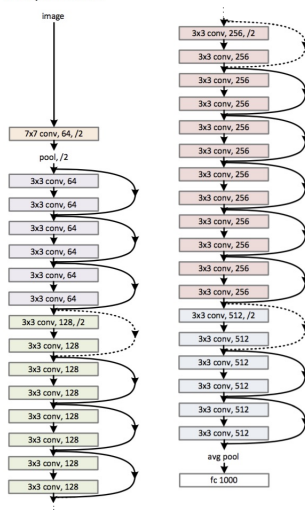




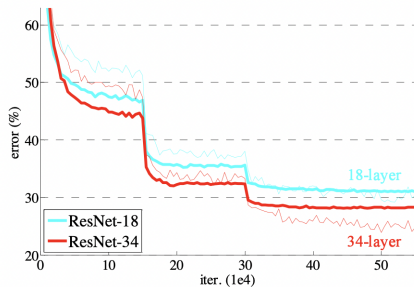
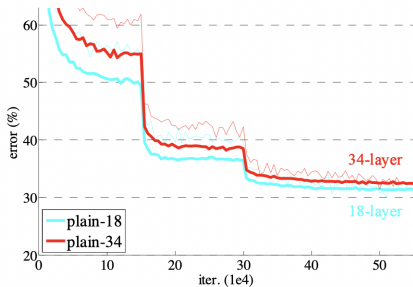
Identity function is easy to learn for residual block.

Resnet-50 (2015)

34-layer residual



Resnet-50 (2015)



Resnet-50 (2015)

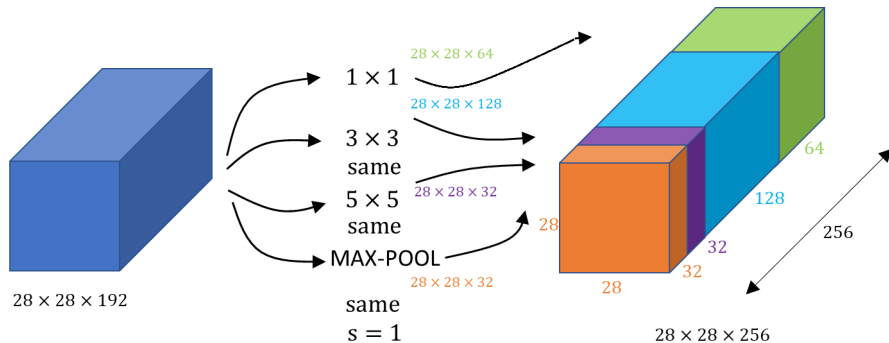
- Resnet-50 has approximately 25.6M parameters.

Resnet-50 (2015)

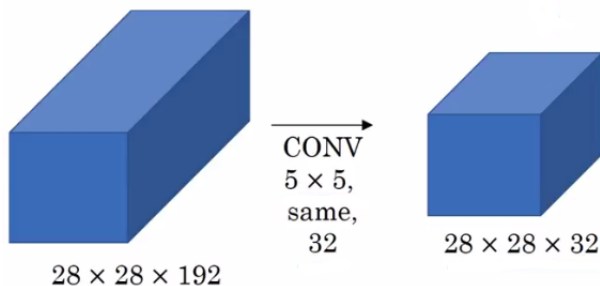
- Resnet-50 has approximately 25.6M parameters.
- Accuracies on ImageNet: Top1=77.15%, Top5=93.29%.

GoogLeNet/Inception v1 (2014)

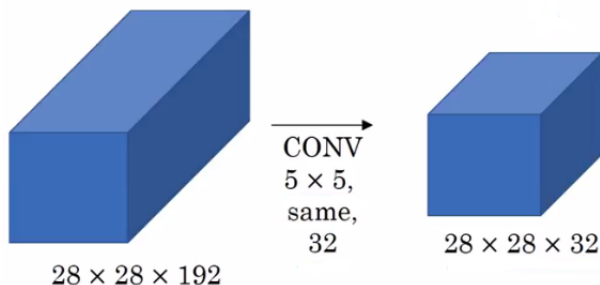
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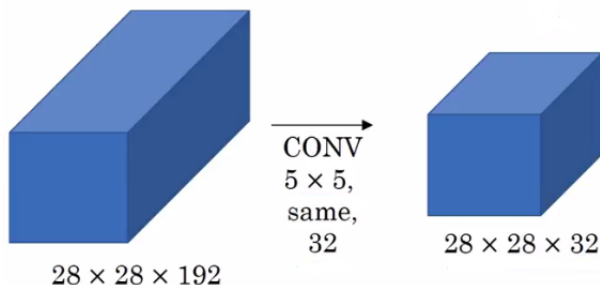
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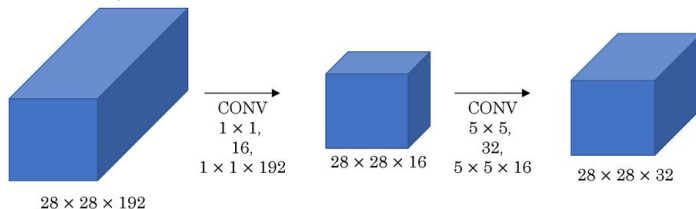
How much is the number of multiplications?



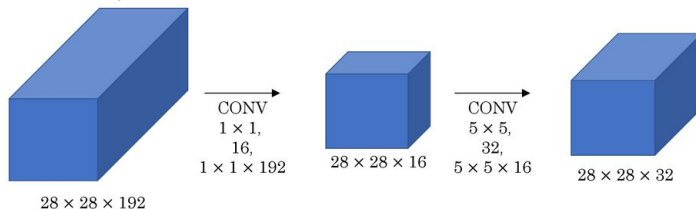
How much is the number of multiplications?

$$28 \cdot 28 \cdot 32 \cdot 5 \cdot 5 \cdot 192 \approx 120M$$

Inception Networks with 1X1 Convolutions

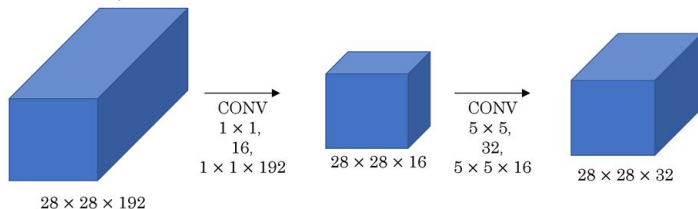


Inception Networks with 1X1 Convolutions



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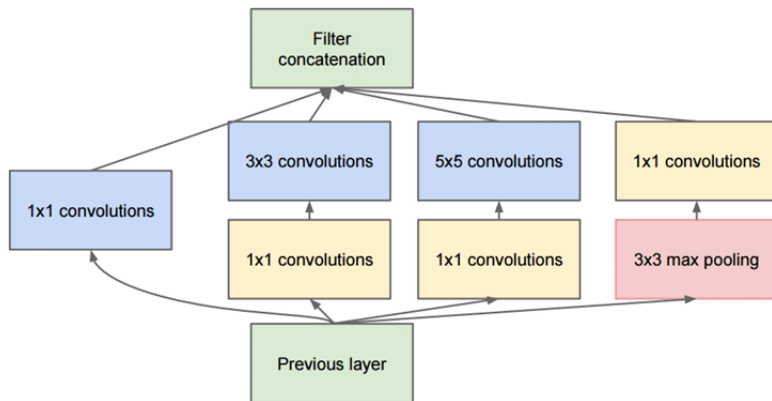
Inception Networks with 1X1 Convolutions



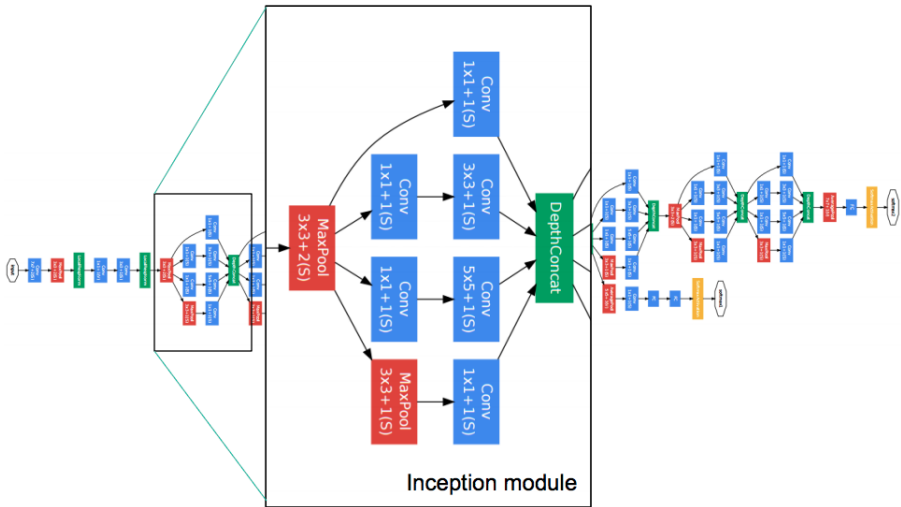
How much is the number of multiplications?

$$28 \cdot 28 \cdot 16 \cdot 1 \cdot 1 \cdot 192 + 28 \cdot 28 \cdot 32 \cdot 5 \cdot 5 \cdot 16 \approx 12.4M$$

GoogLeNet/Inception v1 (2014)



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Receptive Field

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The Receptive Field of a layer F_k with respect to the layer F_m ($m < k$) is the maximal region in the layer F_m , each element of which is contributed in forming one of pixels in F_k , considering only convolutional layers as contribution.

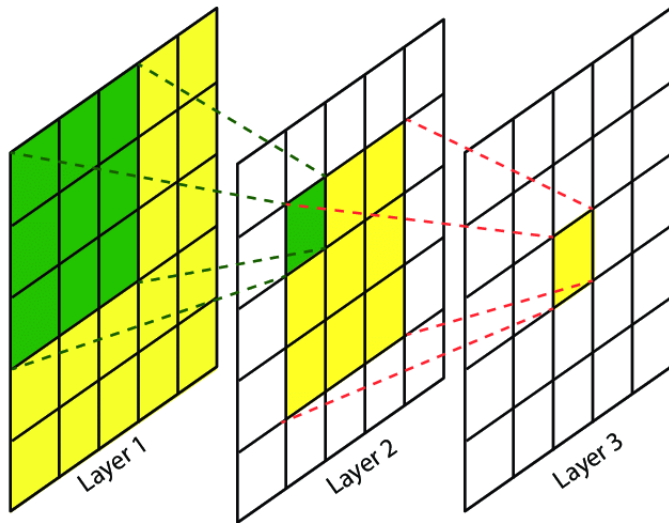
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The Receptive Field of a layer F_k with respect to the layer F_m ($m < k$) is the maximal region in the layer F_m , each element of which is contributed in forming one of pixels in F_k , considering only convolutional layers as contribution. By receptive field of a network we mean the receptive field of its output with respect to its input.

Receptive Field



Remark 1

Although the receptive field can be referred as the "area under vision of the net", not all pixels in receptive field are "equally contributed" in formation of an output pixel. So there is a concept of effective receptive field.

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So the natural need arises to enlarge the size of the receptive field of the network, or keep the size of receptive field the same but decrease the number of computations. For the latter purpose the approach of convolution factorization is appropriate.

Convolution Factorization

The concept of convolution factorization was introduced in the Inception-V3 architecture.

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- replace 5×5 convolution layers with two 3×3 convolution layers,
- replace $n \times n$ convolution layers with asymmetric consequent convolution layers of sizes $n \times 1$ and $1 \times n$.

Convolution Factorization

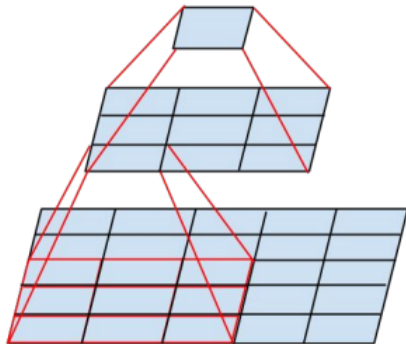


Figure 1. Mini-network replacing the 5×5 convolutions.

Convolution Factorization

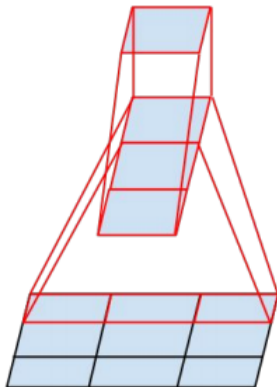


Figure 3. Mini-network replacing the 3×3 convolutions. The lower layer of this network consists of a 3×1 convolution with 3 output units.